CHALMERS

Finite Element Method - Structures, TME245

Computer assignment 2: Plate analysis

Instructions

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TME245 FEM - Structures 2024/2025

Preface

Within this computer exercise, a wall section of a swimming pool, subjected to water pressure and an external edge traction, is to be analysed. The tasks will involve to implement a quadrilateral Kirchhoff plate element, to use that to analyse the deformations and stress distribution in the wall section, and then finally to analyse the risk of buckling using second order Kirchhoff theory.
The following MATLAB-files are made available to support this computer exercise:

- `rectMEsh.m`
  A function file that generates the mesh of the plate, and provides the relevant matrices `Coord`, `Edof_ip` and `Edof_ip`

Learning outcomes: The learning outcomes of this project is to learn to implement a Kirchhoff plate element, to learn how to expand a standard Kirchhoff plate element to include second order in-plane effects (to allow for a linearise buckling analysis) and to conduct finite element simulations of plate problems, including post-processing of displacements and stresses.

Göteborg in February 2025

Martin Fagerström

Problem definition

A designer wants to design and build her own swimming pool. For simplicity, she is planning to make it from rectangular metals sheets welded together, see Figure 1. Furthermore, she aims to reinforce the walls by welding additional beams onto the metal sheets, equally spaced along each wall, and then connecting those beams to the ground (see red vertical lines in Figure 1). On the top of one of the longer sides, she will also mount a platform to allow people to enter and leave the pool.

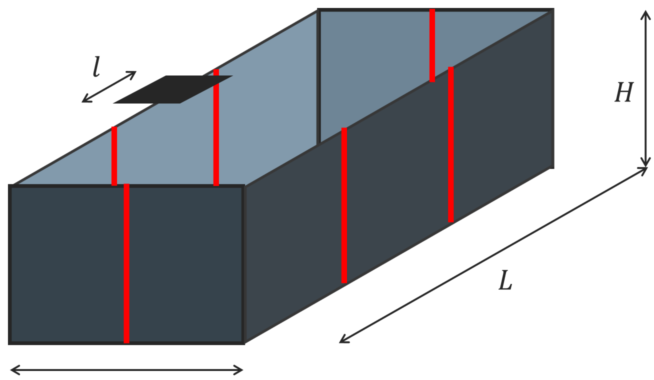

Figure 1: Swimming pool geometry to be considered in Computer Assignment 2. Red vertical lines indicate the location of reinforcement beams, equally spaced along each side.

The dimensions of the pool are $L = 6\,\mathrm{m}$, $W = 4\,\mathrm{m}$ and $H = 2\,\mathrm{m}$, and the platform (width $l = 1\,\mathrm{m}$) is placed at the centre of the upper edge of one of the longer sides. Furthermore, the pool is to be made from steel with the following properties: Young's modulus $E = 210\,\mathrm{GPa}$, Poisson's ratio $\nu = 0.3$ and yield stress $\sigma_{\text{yield}} = 450$ MPa.

The design requirements are such that the safety factor against initial yielding or buckling (whichever is most critical) should be at least 2, considering combined loads from the water pressure (assuming the water line to reach the top of the pool), and a person of 150 kg standing on the platform (assume the load evenly distributed along l). The task for the designer is then to find the minimum steel sheet thickness that fulfills both requirements. Your job, however, is to analyse the safety factor against yielding (Task 1) and buckling (Task 2), considering a sheet thickness of $t = 5\,\mathrm{mm}$

The most critical part of the pool will be the long side where the platform is mounted. To clarify the boundary conditions, the surface of the pool side to be analysed is subjected to a pressure that increases linearly with the pool depth $p(y) = \rho g y$, see left part of Figure 2. Furthermore, the load from the person standing on the platform is to be considered as a distributed traction $\bar{t}$ (in the vertical direction) along l. Finally, as a simplification, each segment between reinforcement beams are considered individually, with the vertical and bottom sides assumed as clamped. To avoid confusion, the upper edge is to be considered as free to move.

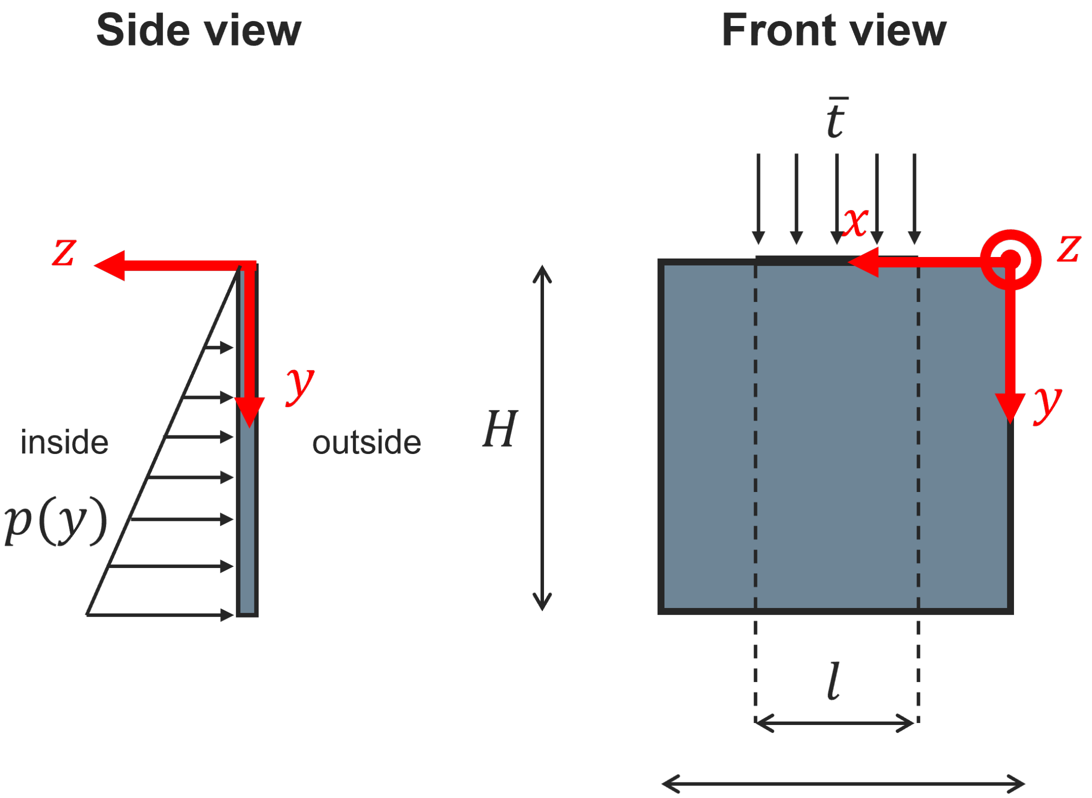

Figure 2: Side and front view of the pool wall section to be analysed in Computer Assignment 2. Left, right and bottom side are assumed to be clamped.

Task 1: Linear elastic plate analysis using MATLAB or Python

Complete the following tasks and comment on the results when appropriate.

a) **Make your own sketch of the problem, where you clearly indicate the coordinate system and the boundary conditions**. Furthermore, **state (don't derive) the weak form of the problem, and make sure to explicitly explain how the boundary conditions enter in the formulation**. **(0.5p)**

b) **Describe the implementation procedure for a Kirchhoff plate element that considers both in-plane and out-of-plane loading and that outputs the total element stiffness matrix and the element external force contributions resulting from the pressure load**. Define any matrices you introduce.

Then, **implement the element as a MATLAB- or Python-function file to be used to solve the problem**. To simplify the implementation, please approximate the pressure to be constant over each element, with the magnitude given by $p_e = \rho g \overline{y}_e$ where $\overline{y}_e$ is the vertical position (measured from the top) of the midpoint of the element (found by the mean value of the nodal $y-$coordinates). **Check and report that the sum of all nodal force contributions equals the total force generated by both the pressure load and the load from the 150 kg person on the platform.**

Please also note that code snippets 26, 27 and 30 may be helpful in completing this task. But note that the code in snippet 30 needs to be extended to also include the in-plane behaviour.

Make sure to compare your element routine output, $\underline{K}_{uu}$, $\underline{K}_{ww}^{K,e}$ and $\underline{f}^{e,ext}$, against the test case provided on Canvas. **(1.0p)**

c) **Write a small FE-program for plate analysis to solve the problem with the combined loads from the water pressure and the traction** $\overline{t}$. Please note that the in-plane traction load is computed identically to the water pressure load in Computer Exercise 1.

To generate the mesh, please use the provided MATLAB-function `rectMesh` which generates the necessary matrices `Coord`, `Edof_ip` and `Edof_oop`, for which the degrees of freedom per node are ordered as $[u_x \ u_y]$ for `Edof_ip` and $[w \ \theta_x \ \theta_y]$ for `Edof_oop`.

Here, it may be helpful to look at code snippets 28 and 29, but again with the consideration that the in-plane behaviour needs to be added.

Create a plot of the deformed shape of the plate, and report the maximum in-plane and out-of-plane displacement in terms of locations (nodal coordinates) and values. (1.0p)

d) **Implement a MATLAB or Python function that computes the in-plane stress components in the element integration points ($\xi_1 = \pm\frac{1}{\sqrt{3}}; \xi_2 \pm \frac{1}{\sqrt{3}}$ in the parent domain) for a given thickness coordinate.**
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Use this function to **create three contour plots where, for each plot, each plate element is coloured by the in-plane average of the von Mises effective stress** (make sure to first average the stress components over the element and then compute the resulting von Mises stress). The plots should be created for the three different thickness coordinates $z = -h/2$, $z = 0$ and $z = h/2$, where $h$is the thickness of the plate.

Please note that, in Voigt form, the von Mises stress σ_{vM} is found by:

$$\underline{s} = \begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{12} \\ \sigma_{13} \\ \sigma_{23} \end{bmatrix} - \frac{(\sigma_{11} + \sigma_{22} + \sigma_{33})}{3} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \sigma_{\text{vM}} = \sqrt{\frac{3}{2}\left(\underline{s}^T \underline{s} + \sigma_{12}^2 + \sigma_{13}^2 + \sigma_{23}^2\right)}.$$

(1.0p)

Summary of things to report

- A sketch of the problem, with clear definitions of the boundary conditions (subtask a))
- A complete statement of the weak form of the problem (subtask a))
- A description of the implementation procedure for a Kirchhoff plate element (subtask b))
- MATLAB or Python code for Kirchhoff plate element (subtask b))
- A check of the accuracy of the external force vector (subtask b))
- A plot of the deformed plate, together with the location of values and locations of maximum in-plane and out-of-plane displacements, respectively (subtask c))
- A code solving the problem (subtask c))

- A code computing the stresses in a Kirchhoff element (subtask d))
- 3 contour plots of the per element in-plane averaged von Mises stress at the locations through the plate thickness (subtask d))

Task 2: Linearised buckling using MATLAB or Python

The next task is to compute the safety factor against buckling given the in-plane plate load coming from a 150 kg person standing on the platform. As such, the following tasks should be completed:

a) Starting from your element function from Task 1, **implement an element function (in MATLAB or Python) for a Kirchhoff element to be used for linearised prebuckling analysis**. For this subtask, you may assume that the $\underline{\tilde{N}}^{\mathrm{sec(R)}}$ is given as a constant input to the element. This, the task that remains is to implement the computation of $\underline{G}^{\mathrm{e,(R)}}$ using a 3x3 Gauss integration scheme.

To clarify, the new function file should have material properties, nodal coordinates, the thickness and a constant element $\underline{\tilde{N}}^{\mathrm{sec(R)}}$ matrix as input, and have (at least) $\underline{K}_{ww}^{K,e}$ and $\underline{G}^{\mathrm{e,(R)}}$ as output.

Describe the main steps of your implementation in the report.

(1.0p)

b) **Implement a small finite element programme for linearised prebuckling analysis** that includes the following steps:

1. A solution of the in-plane deformation problem with the in-plane load coming from the distributed traction $\bar{t}$. Here, you need to assemble the in-plane element stiffnesses $\underline{K}_{uu}^{e}$ into $\underline{K}_{uu}$, add the contribution of the external in-plane traction into $\underline{f}_{u}^{ext}$ and then solve

$$\underline{K}_{uu}\,\underline{a}_u = \underline{f}_u^{ext}$$

with the proper in-plane displacement boundary conditions.

Note that all you need should have been implemented as part of solving Task 1.

Please note that if you implement the applied traction as that resulting from the 150 kg load, the smallest eigenvalue of the buckling problem (see step 4 below) will correspond to the safety factor against buckling.

2. For each element, evaluate the in-plane stress components (use your code from Task 1) in the element midpoint (in-plane midpoint and at $z = 0$) and compute

$$\underline{\tilde{N}}^{\mathrm{e,sec(R)}} \approx h \begin{bmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{xy} & \sigma_{yy} \end{bmatrix}$$

(h is the plate thickness) as an approximation of the in-plane loading that may trigger buckling. Then use this matrix as an element representative in-plane load[1] in the function file developed in Task 2a.

3. Use the function file created in Task 2a, with $\tilde{\underline{N}}^{\mathrm{e,sec(R)}}$ computed as described in the previous step, to compute $\underline{K}_{ww}^{K,e}$ and $\underline{G}^{\mathrm{e,(R)}}$ for each element. Assemble these contributions into $\underline{K}_{ww}^{K}$ and $\underline{G}^{\mathrm{(R)}}$. **Make sure to verify also the calculation of $\underline{G}^{\mathrm{(R)}}$ against the test case provided on Canvas.**
4. Find the smallest eigenvalue λ_1 and the corresponding buckling mode $\underline{z}_1$ from the generalised eigenvalue problem

$$\underline{K}_{wwFF}^{K}\underline{z}_i = -\lambda_i \underline{G}_{FF}^{\mathrm{(R)}}\underline{z}_i$$

where $\underline{K}_{wwFF}^{K}$ and $\underline{G}_{FF}^{\mathrm{(R)}}$ are the parts of $\underline{K}_{ww}^{K}$ and $\underline{G}^{\mathrm{(R)}}$ associated with the free degrees of freedom.

Use this programme to **calculate and report the safety factor against buckling of the pool wall. Also create and report a plot of the shape of the most critical buckling mode**. For the latter, plot the deformed shape using the corresponding out-of-plane degrees of freedom, this disregarding the degrees of freedom that define rotations. **(2.5p)**

Summary of things to report

- MATLAB or Python code for a Kirchhoff plate element for linearlised pre-buckling analysis (subtask a))
- A description of the main steps of the implementation of the element (subtask a))
- The code to solve subtask b)
- Report the safet factor against buckling and explain how it has been computed (subtask b))
- A plot of the most critical buckling mode (subtask b)

[1]Please note that, in general, $\tilde{\underline{N}}^{\mathrm{sec(R)}}$ will vary inside the element. However, as this variation is often rather small we simplify the analysis by considering the midpoint resultants as representative constant values for the element.